Calculate gravitational force

antocciphysics | TEACHER KEY | CPA

Learning target: Calculate a gravitational interaction with consistent units.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 2; 3 course days later.*
30 m/s downward.

02. Explain a worked example (worked)
Opposite attractive 2 N arrows; N·m² divided by m² gives N.

03. Look and explain (model)
The required separation is 4 m, not either half-distance.

04. Build the solution (practice)
r²16 m²; F 1 N; at 8 mF=.25 N.

05. Use the idea (reason)
Small G makes force tiny for modest masses; inability to feel a force does not make it zero.

06. Choose a method (mixed) *Review: Unit 1, Day 15; 12 course days later.*
Only weight 40 N downward from Earth. The hand is no longer in contact; upward velocity is not a force.

07. Exit check (exit)
4 N; attractive toward the other body.

08. STRETCH: Apply it another way (stretch)
At 6 m: 0.5 N. For 18 N, r²=1 m² so r=1 m.

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Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 2; 3 course days later.*
v_f=23 m/s down; Δy=26 m down.

02. Explain a worked example (worked)
Opposite attractive 2 N arrows; N·m² divided by m² gives N.

03. Represent and interpret (model)
The required separation is 4 m, not either half-distance.

04. Solve and justify (practice)
F=.04002 N≈.0400 N; at 20 m=.0100 N.

05. Analyze the reasoning (reason)
Small G makes force tiny for modest masses; inability to feel a force does not make it zero.

06. Choose a method (mixed) *Review: Unit 1, Day 15; 12 course days later.*
Weight down, normal up, kinetic friction right. No vertical acceleration; rightward horizontal acceleration if friction is the only horizontal force.

07. Exit check (exit)
1.6675×10^{-5} N; wrong denominator 2 instead 4 makes answer twice too large.

08. STRETCH: Apply it another way (stretch)
Product 0.48 N·m²; r=√(0.48/0.03)=4 m.